# Revision_Deflation — the 16-component Dirac field and its singlet superfluid on spacetimes whose coordinates x5, x6, x7 deflate exponentially

PROJECT spacetimes: where to start, what was found, and what is open

Prepared for Patrick Lee Nash, Ph.D.

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## What this folder is

The main project (the folder above this one) studies a 16-component Dirac field on two flat eight-dimensional spacetimes, of signatures 4+4 and 7+1, and a superfluid of fermion pairs bound by a scalar field, the Vorton-Scalar. On November 12, 2026 the author paused the main project and instructed that every calculation be repeated under new hypotheses:

> the coordinates $x5$, $x6$, $x7$ deflate exponentially; Minkowski space is replaced by a pseudo-Riemannian space with a diagonal metric whose components along these coordinates are $g_{55} = -\text{const}^2 e^{-2\,\text{g55}\,x4}$, and likewise $g_{66}$ with the rate g66 and $g_{77}$ with the rate g77.

Here g55, g66, g77 are the author's names for three positive rates, and $g_{55}$, $g_{66}$, $g_{77}$ are components of the metric. This folder contains the repetition: programs, data, figures, documents and two manuscripts. Nothing in the folder above was changed, except one line of a build script (recorded in the provenance P11 of the main project).

## Where to start

| if you want | read |
|---|---|
| the two papers | `manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 4+4 Deflating Spacetime.pdf` and `... on 7+1 Deflating Spacetime.pdf` |
| the geometry and the Dirac equations, derived step by step | `dirac matrices deflating.pdf` (DMD) |
| the superfluid on the deflating spacetimes | `supplement/RS1_vorton_scalar_and_superfluid_deflating.pdf` |
| the effective action along the deflation | `supplement/RS4_effective_action_along_the_clock.pdf` |
| the Higgs hypothesis | `supplement/RS2_higgs_hypothesis_deflating.pdf` |
| cosmology, dark matter, the Hubble tension | `supplement/RS3_cosmology_deflating.pdf` |
| the microwave background and the growth of structure | `supplement/RS5_microwave_background_deflating.pdf` |
| what was assumed, what mirrors what, what is open | `provenance/RP0_hypotheses_and_status.pdf` |
| the proof that no symbol has two meanings | `provenance/RP1_master_symbols_deflation.pdf` |
| how to run everything | `provenance/RP8_code_deflation.pdf` |

| if you want | read |
|---|---|
| every error that was made | `provenance/RP9_error_log_deflation.pdf` |

Every document exists as `.md`, `.tex` and `.pdf`.

# What was found

**Geometry** (DMD, Part 1). The deflating spacetimes are homogeneous, with 14, 15 or 20 symmetries (three different, two equal, three equal rates) instead of the 36 of flat spacetime. They are not vacuum solutions of Einstein's equations. *With three equal rates they are exact vacuum solutions of Einstein–Gauss–Bonnet gravity*, for a cosmological constant $3\,\mathrm{g55}^2$ and a Gauss–Bonnet coupling $-1/(4\,\mathrm{g55}^2)$.

**The Dirac field** (DMD, Parts 2 to 4). A field that does not depend on $x5, x6, x7$ obeys, after a rescaling by the square root of the deflating volume, exactly the flat equations. A field that does depend on them behaves in opposite ways on the two spacetimes. On **7+1** the deflation creates particles, with a Fermi–Dirac spectrum of the temperature rate$/2\pi$. On **4+4** every admissible mode that depends on a deflating time is a standing wave that is extinguished as the exponential of an exponential: *the deflation removes the dependence on the additional times, which the main project had to impose by hand.*

**The superfluid** (RS1, RS4). The model of the main project holds unchanged, with a coupling that grows: $\alpha_{VS} \propto e^{\tau_d}$, where $\tau_d = \mathrm{gsum}\, x4$ is the number of e-foldings of the deflating volume; the masses are constant. Every threshold of the main project is therefore passed at a definite time. The gap grows much faster than the coupling, $d\ln(\mathrm{gap})/d\tau_d = 1/\lambda$. Two fermions become bound at a scattering resonance. The paired state can follow the deflation only once the gap exceeds a few times gsum$/\lambda$; before that it lags and breaks pairs. A vortex line carries four fermion zero modes, two for each direction.

**The Higgs hypothesis** (RS2). The internal symmetry that gives the pair condensate the pattern of the Higgs vacuum survives exactly, but on the deflating 7+1 spacetime it is no longer a symmetry of the spacetime. And if the condensate were the Higgs field, the electroweak scale would follow the growing coupling; atomic clocks then limit the rate of deflation to about $10^{-6}$ of the Hubble constant. *The Higgs hypothesis and a deflation of any consequence today exclude each other.*

**Cosmology** (RS3). If gravity is the eight-dimensional Einstein theory, Newton's constant grows with the deflation, lunar laser ranging limits the rate to $3 \times 10^{-3}$ of the Hubble constant, and the Hubble constant inferred from the microwave background *falls*. The Hubble tension is not relieved. For dark matter the deflation adds an epoch of resonant self-interaction; the conclusions of the main project about the masses are unchanged.

**The microwave background and the growth of structure** (RS5). The spectra need one assumption more: that the scale factors of $x5, x6, x7$ are not perturbed when ordinary space is. Even then the Einstein equations of ordinary space cannot all hold with matter alone. Whatever keeps the three rates constant must carry a tension and perturbations of it, and the hypotheses do not say which. *What does not depend on that choice:* the spectra at the multipoles $\ell \geq 30$. Mock data of the quality of the Planck satellite would bound the rate to $0.0385\, H_0$, between the bounds of lunar laser ranging and of nucleosynthesis; in every fit the Hubble constant falls, by 84 km/s/Mpc per unit of gsum$/H_0$; and the rate cannot help the dark radiation of the main project to raise the Hubble constant. *What does depend on it:* the growth of structure and the largest angular scales, which the revision brackets with two choices.

## What is open

1. What keeps the three rates constant. The revision calls it the agent and assumes nothing about its nature. The spectra of the microwave background were computed under an assumption about it that the author is asked to confirm or replace (RP0, Sec. 5); the growth of structure remains undetermined until it is specified.
2. The stability of the Einstein–Gauss–Bonnet solution, and the source of the deflation for unequal rates.
3. The back-reaction of the created particles (7+1), and a principle that excludes the growing solutions (4+4).
4. The end of every history: the couplings grow without bound, and the model fails when they become strong.

The stages of the main project that were paused are listed in its provenance P11 and remain paused.

## Numbers

The Rust program `deflation` performs 238 checks and has 49 unit tests; an independent Python script performs 43 of 43 checks; the registry of symbols 6 of 6. All pass. A further script compares every number that is quoted in the running text of the documents and of the manuscripts with the numbers that the programs state (974 numbers in 29 documents, 0 not traced). The folder contains 19 figures and the data files from which every table and figure is made.

## How to run it

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation"
python3 code/scripts/vendor_eb.py --check
cargo build --release --manifest-path code/deflation-rs/Cargo.toml
cargo test  --release --manifest-path code/deflation-rs/Cargo.toml
# the first nine commands: about three and a half minutes together
for c in matrices modes history quench cosmology symmetry channels matter paired; do
  ./code/deflation-rs/target/release/deflation $c
done
# the Einstein-Boltzmann calculations of RS5: about an hour and ten minutes together
for c in background spectrum forecast families superfluid; do
  ./code/deflation-rs/target/release/deflation $c
done
python3 code/scripts/verify_deflating_independent.py
```

The provenance RP8 explains every step, and the remaining ones (symbols, tables, figures, documents).